

Proximity to Nuclear Reactors and Household Income in U.S. Census Tracts

A Spatial Analysis of over 80,000 Tracts

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```
##
## Attaching package: 'lubridate'
```

```
## The following objects are masked from 'package:base':
##
##   date, intersect, setdiff, union
```

```
## here() starts at C:/Users/lexcl/OneDrive/Documents/EDE/EDE_Spring2026
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr   1.2.0     ✓ readr   2.1.5
## ✓ forcats 1.0.0     ✓ stringr 1.6.0
## ✓ ggplot2 4.0.0     ✓ tibble  3.3.1
## ✓ purrr   1.1.0     ✓ tidyr   1.3.2
## — Conflicts — tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()     masks stats::lag()
## ⓘ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
##
## Attaching package: 'janitor'
##
##
## The following objects are masked from 'package:stats':
##
##   chisq.test, fisher.test
```

```
## Warning: package 'tidycensus' was built under R version 4.5.3
```

```
## Linking to GEOS 3.13.1, GDAL 3.11.4, PROJ 9.7.0; sf_use_s2() is TRUE
```

```
## Warning: package 'modelsummary' was built under R version 4.5.3
```

```
##
## Attaching package: 'modelsummary'
##
## The following object is masked from 'package:tidycensus':
##
##   get_estimates
```

DATA WRANGLING STEP 1: Joining reactor coordinate and water source data

#I am working with two datasets to get the lat and long of each site. In some instances, like at McGuire, there are multiple units at one site. In this project I will be working with distance from sites, not individual units.

```
gen_raw <- read.csv(here("Final_Project/Data/nuclear_units.csv"), skip=1, check.names = FALSE)

gen <- gen_raw %>% clean_names()

gen$nameplate_capacity_mw <- as.numeric(gsub(",", "", gen$nameplate_capacity_mw))

#filtering to nuclear units

nuc_units <- gen %>%
  filter(technology=="Nuclear") %>%
  select(utility_id, utility_name, plant_code, plant_name, state, county, nameplate_capacity_mw, operating_year)

#aggregating units to sites by plant codes.

nuc_sites <- nuc_units %>%
  group_by(plant_code) %>%
  summarise(
    n_units = n(),
    total_nameplate_mw = sum(nameplate_capacity_mw, na.rm = TRUE),
    first_online_year = min(operating_year, na.rm = TRUE),
    last_online_year = max(operating_year, na.rm = TRUE),
    .groups = "drop"
  )
```

```
plant.raw <- read.csv(here("Final_Project/Data/plant_watersource.csv"), skip = 1, check.names = FALSE)

plant <- plant.raw %>% clean_names()
```

```
nuclear_join <- plant %>%
  inner_join(nuc_sites, by="plant_code") %>%
  select(plant_code, plant_name, utility_name, state, county,
    latitude, longitude, water_source = name_of_water_source,
    n_units, total_nameplate_mw,
    first_online_year, last_online_year)

write.csv(nuclear_join,
  here("Final_Project/Data/Processed/nuclear_join.csv"),
  row.names = FALSE)
```

DATA WRANGLING STEP 2: Pulling Household Income with Tidycensus

```

conus_states <- state.abb[!state.abb %in% c("AK","HI")]

acs_tracts <- get_acs(
  geography = "tract",
  variables = "B19013_001",
  year = 2024,
  survey = "acs5",
  state = conus_states,
  geometry = TRUE,
  output = "wide"
) %>%
  rename(median_hh_income = B19013_001E,
         median_hh_income_moe = B19013_001M) %>%
  st_transform(5070)

saveRDS(acs_tracts, here("Final_Project/Data/Processed/acs_tracts.rds"))

```

```

acs_tracts <- read_rds(here("Final_Project/Data/Processed/acs_tracts.rds"))

```

```

nuclear_sf <- nuclear_join %>%
  st_as_sf(coords = c("longitude", "latitude"), crs=4326, remove = FALSE)

nuclear_albers <- nuclear_sf %>% st_transform(5070)

saveRDS(nuclear_sf,
        here("Final_Project/Data/Processed/nuclear_sf.rds"))

```

```

tract_centroids <- st_centroid(acs_tracts)

```

```

## Warning: st_centroid assumes attributes are constant over geometries

```

```

nearest_idx <- st_nearest_feature(tract_centroids, nuclear_albers)

tracts_with_dist <- acs_tracts %>%
  mutate(
    nearest_plant_code = nuclear_albers$plant_code[nearest_idx],
    nearest_plant_name = nuclear_albers$plant_name[nearest_idx],
    dist_to_reactor_m = as.numeric(
      st_distance(tract_centroids, nuclear_albers[nearest_idx, ], by_element = TRUE)
    ),
    dist_to_reactor_mi = dist_to_reactor_m / 1609.34
  )

#count NA

count_na <- tracts_with_dist %>%
  filter(is.na(dist_to_reactor_m) | is.na(median_hh_income)) %>%
  nrow()

count_na

```

```
## [1] 1435
```

#dropping NA for track distance and mm hh income

```
tracts_clean <- tracts_with_dist %>%  
  filter(!is.na(dist_to_reactor_m),  
         !is.na(median_hh_income))  
  
saveRDS(tracts_clean, here("Final_Project/Data/Processed/tracts_clean.rds"))
```

#For this analysis part, I decided to create bins based off of established NRC planning zones. The NRC defines the first EPZ as the plume exposure pathway extending about 10 miles in radius around the reactor site. The plume exposure pathway extends about 50 miles in radius. Since these are rather large radii, I created subgroups and then bins from each subgroup. For each subgroup, n >= 30.

#Since only 3 census tracts fell within 1 mi of a nuclear site, I used 0-5 miles as the first bin, where n = 193, to have a large enough sample size of tracts.

```
bin_samplesize_check <- tracts_clean %>%  
  st_drop_geometry() %>%  
  summarise(  
    under_1 = sum(dist_to_reactor_mi < 1),  
    under_5 = sum(dist_to_reactor_mi < 5),  
    under_10 = sum(dist_to_reactor_mi < 10)  
  )
```

#I then used the cut() function to create a new column called dist_bin that cuts the dist_to_reactor_mi column based on a vector of specific distance ranges. <https://www.statology.org/cut-function-in-r/>

```
tracts_clean_binned <- tracts_clean %>%  
  mutate(  
    dist_bin = cut(dist_to_reactor_mi,  
                   breaks = c(0, 5, 10, 50, Inf),  
                   labels = c("0-5 mi (inner Plume EPZ)",  
                              "5-10 mi (outer Plume EPZ)",  
                              "10-50 mi (Ingestion EPZ)",  
                              "50+ mi (outside EPZs)"),  
                   include.lowest = TRUE),  
    state_fips = substr(GEOID, 1, 2),  
    log_income = log(median_hh_income)  
  )  
  
tracts_clean_binned$dist_bin <- relevel(factor(tracts_clean_binned$dist_bin),  
                                       ref = "50+ mi (outside EPZs)")
```

```
tract_lm <- st_drop_geometry(tracts_clean_binned)  
  
lm.model.one <- lm(log_income ~ dist_bin + factor(state_fips), data = tract_lm)  
  
model_one_summary <- summary(lm.model.one)  
  
model_one_summary
```

```
modelsum <- modelsummary(lm.model.one,
  coef_omit = "state_fips|Intercept",
  stars = TRUE)

modelsum
```

#Now I will do a Tukey-HSD test. Since it allows for "all possible pairwise comparisons" which will allow me to do a bin-to-bin comparison of household income as a function of reactor proximity. <https://www.r-bloggers.com/2021/08/how-to-perform-tukey-hsd-test-in-r/>

```
anova.model <- aov(log_income~dist_bin, data = tract_lm)
```

```
bin_comparison <- TukeyHSD(anova.model)
bin_comparison
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = log_income ~ dist_bin, data = tract_lm)
##
## $dist_bin
##
```

	diff	lwr
## 0-5 mi (inner Plume EPZ)-50+ mi (outside EPZs)	0.140162599	0.054335663
## 5-10 mi (outer Plume EPZ)-50+ mi (outside EPZs)	0.047531157	0.002466009
## 10-50 mi (Ingestion EPZ)-50+ mi (outside EPZs)	0.005498799	-0.004017845
## 5-10 mi (outer Plume EPZ)-0-5 mi (inner Plume EPZ)	-0.092631442	-0.189326541
## 10-50 mi (Ingestion EPZ)-0-5 mi (inner Plume EPZ)	-0.134663800	-0.220743072
## 10-50 mi (Ingestion EPZ)-5-10 mi (outer Plume EPZ)	-0.042032358	-0.087576247

```
##
## upr p adj
## 0-5 mi (inner Plume EPZ)-50+ mi (outside EPZs) 0.225989535 0.0001598
## 5-10 mi (outer Plume EPZ)-50+ mi (outside EPZs) 0.092596306 0.0340456
## 10-50 mi (Ingestion EPZ)-50+ mi (outside EPZs) 0.015015442 0.4469272
## 5-10 mi (outer Plume EPZ)-0-5 mi (inner Plume EPZ) 0.004063657 0.0661559
## 10-50 mi (Ingestion EPZ)-0-5 mi (inner Plume EPZ) -0.048584528 0.0003404
## 10-50 mi (Ingestion EPZ)-5-10 mi (outer Plume EPZ) 0.003511530 0.0827240
```

#In the future, I would like to do a Dunnett comparison so I can control for variation between states, but that is beyond the scope of my current knowledge and time for this project.

Methods

I examined the correlation between proximity to nuclear reactors and tract-level median household income, which I log-transformed to minimize the impacts of right-skewed income data. Using 2020-2024 ACS 5-year estimates for 82,122 CONUS census tracts and plant coordinates from the EIA form 860 (2024), I estimated the distance from each tract centroid to its nearest operating nuclear reactor site, and then classified tracts into bins aligned with the Nuclear Regulatory Commission's Emergency Planning Zones.

Because several reactor sites host multiple generating units (for example, McGuire has two units, Vogtle has four), I aggregated reactor units to the site level using the plant code on the Form EIA-860. This produced 56 unique reactor sites from 96 individual generator units.

Because this analysis focuses on the income levels of those who live in immediate proximity to nuclear reactors, I subdivided the 10-mile Plume Exposure Pathway into two bins. Since only 3 census tracts fell within 1 mi of a nuclear site, I used 0-5 miles as the first bin, where $n = 193$, to have a large enough sample size of tracts.

Results

A log-linear regression of median household income on tracts in the following defined bins: “0-5 mi (inner plume EPZ)”, “5-10 mi (outer plume EPZ)”, “10-50 mi (Ingestion EPZ)”, “50+ mi (outside EPZs)”, shows that tracts within the 10-mile plume exposure pathway had statistically significant higher household incomes than the reference group -tracts beyond 50 miles from any reactor. The effect is strongest in the innermost 0-5 mile ring, where tracts exhibit median incomes 20.4% higher than the reference ($p < .001$ or $p = 3.71e-09$) and attenuates to 11.7% in the 5-10 mile ring ($p < .001$ or $p = 3.31e-11$). Beyond the plume EPZ, the effect dampens sharply. Tracts in the 10-50 mile ingestion pathway EPZ have incomes only 2.4% higher than the reference ($p < .001$ or $p = 2.85e-08$).

To test whether this attenuation is a substantive threshold, I used a Tukey HSD for a bin-to-bin comparison of household income as a function of proximity to nuclear reactors. The analysis confirms that the reactor-proximity effects in tracts in the Ingestion Pathway EPZ are statistically indistinguishable from tracts beyond the 50-mile ring ($p = .45$). While the primary regression detects a 2.4% household income premium in the Ingestion EPZ, the effect does not survive a more conservative pairwise comparison, suggesting that notable income-proximity effects are confined within the 10-mile plume exposure pathway.

The Tukey test also shows that the income-proximity effect between the inner plume and outer plume is marginal ($p = .07$), suggesting that the two bins in the 10-mile plume generally behave as one wealthier area in comparison to farther away bins. Therefore, reactor-proximity effects on wealth appear to be spatially correlated and concentrated to the 10-mile zone, dampening beyond that radius.

Limitations

In the future, I would like to do a Dunnett comparison across bins so I can control for variation between states, but that is beyond the scope of my current knowledge and time for this project. Given that reactors are distributed fairly evenly across 28 states, state-level reactor imbalance is unlikely to throw off the Tukey Test.

The R-squared value indicates that about 12% of variance in tract-log income can be explained by the bins or controlling for state. However, other factors that might influence tract-log income, like proximity to urban areas, are unaccounted for in this analysis.

1,435 census tracts were omitted from the analysis because they were missing either median household income or distance to reactor data, 271 due to invalid geometries that precluded distance calculation, and 1,164 due to suppressed ACS median household income estimates.

Conclusion

This pattern is inconsistent with the environmental-justice narrative observed with other kinds of energy-infrastructure (coal plants, oil refineries), and consistent with the hypothesis that nuclear reactor site selection near bodies of water, coincides with lakeside real estate premiums. This is a correlation not causation. Nuclear reactors do not attract affluent residents, rather bodies of water attract both nuclear reactors and affluent residents. Reactor cooling requirements drive reactor siting near reservoirs and lakes, which independently command real-estate premiums. Civilizations have clustered along the Tigris, the Euphrates, the Nile, and the Mississippi for reasons that have nothing to do with fission. Reactors and millionaires merely came online.